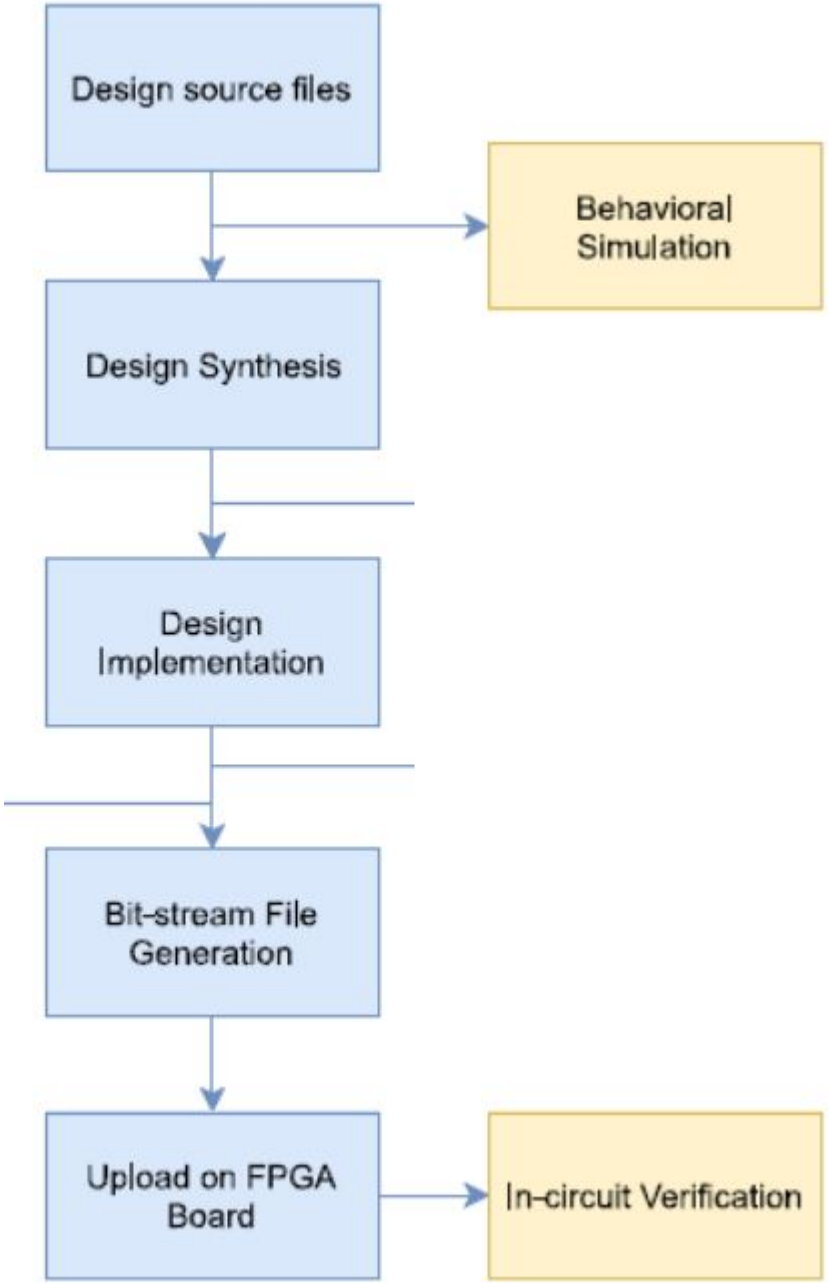
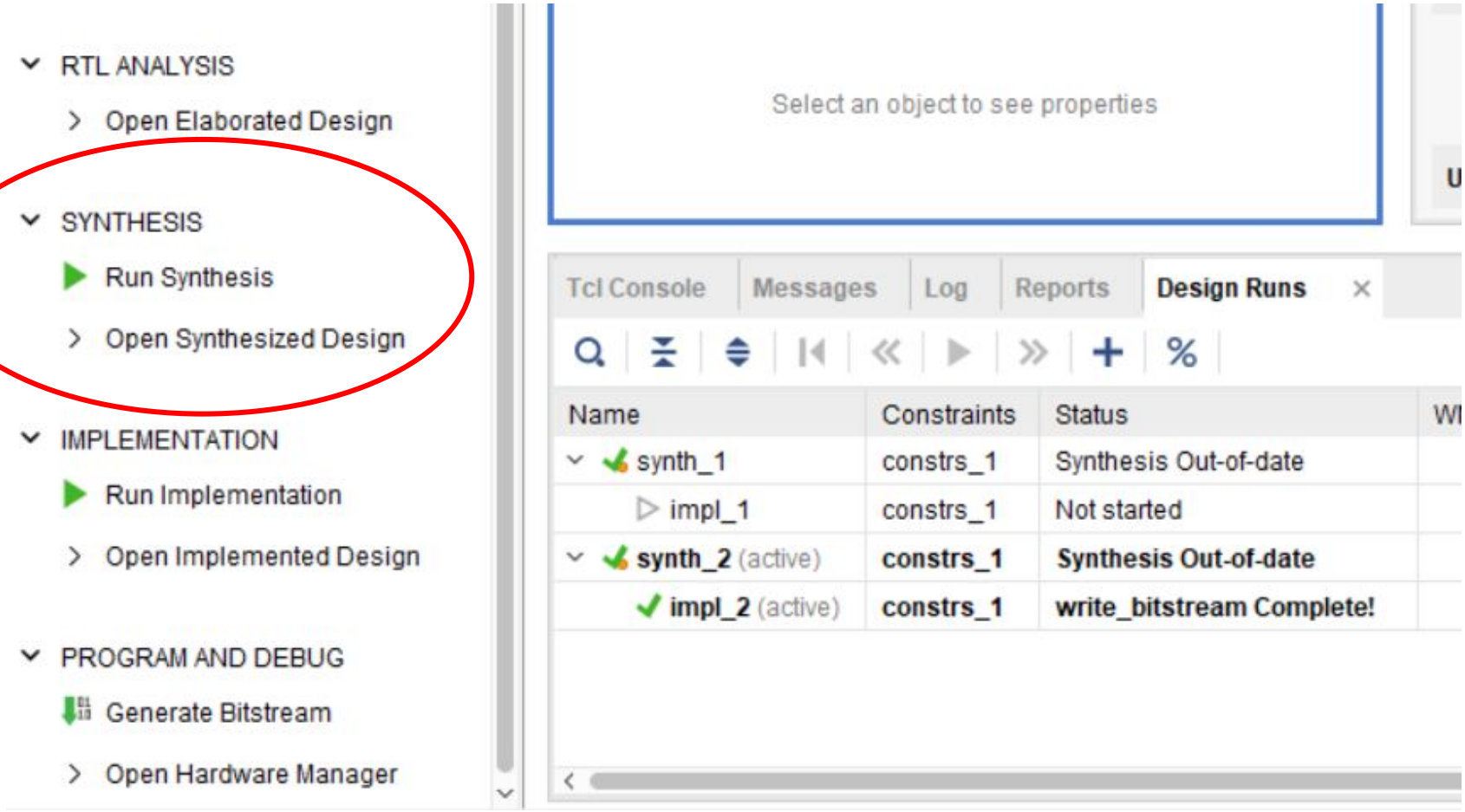
FPGA Design Flow



Ripple Carry Adder design source file (Verilog)

```
module RCA (in0, in1, cin, out, cout);
input [3:0] in0, in1;
input cin;
output [3:0] out;
output cout;
wire c1, c2, c3;
full_adder fa0(in0[0], in1[0], cin, out[0], c1);
full_adder fa1(in0[1], in1[1], c1, out[1], c2);
full_adder fa2(in0[2], in1[2], c2, out[2], c3);
full_adder fa3(in0[3], in1[3], c3, out[3], cout);
endmodule
```

Design Synthesis



Design Implementation

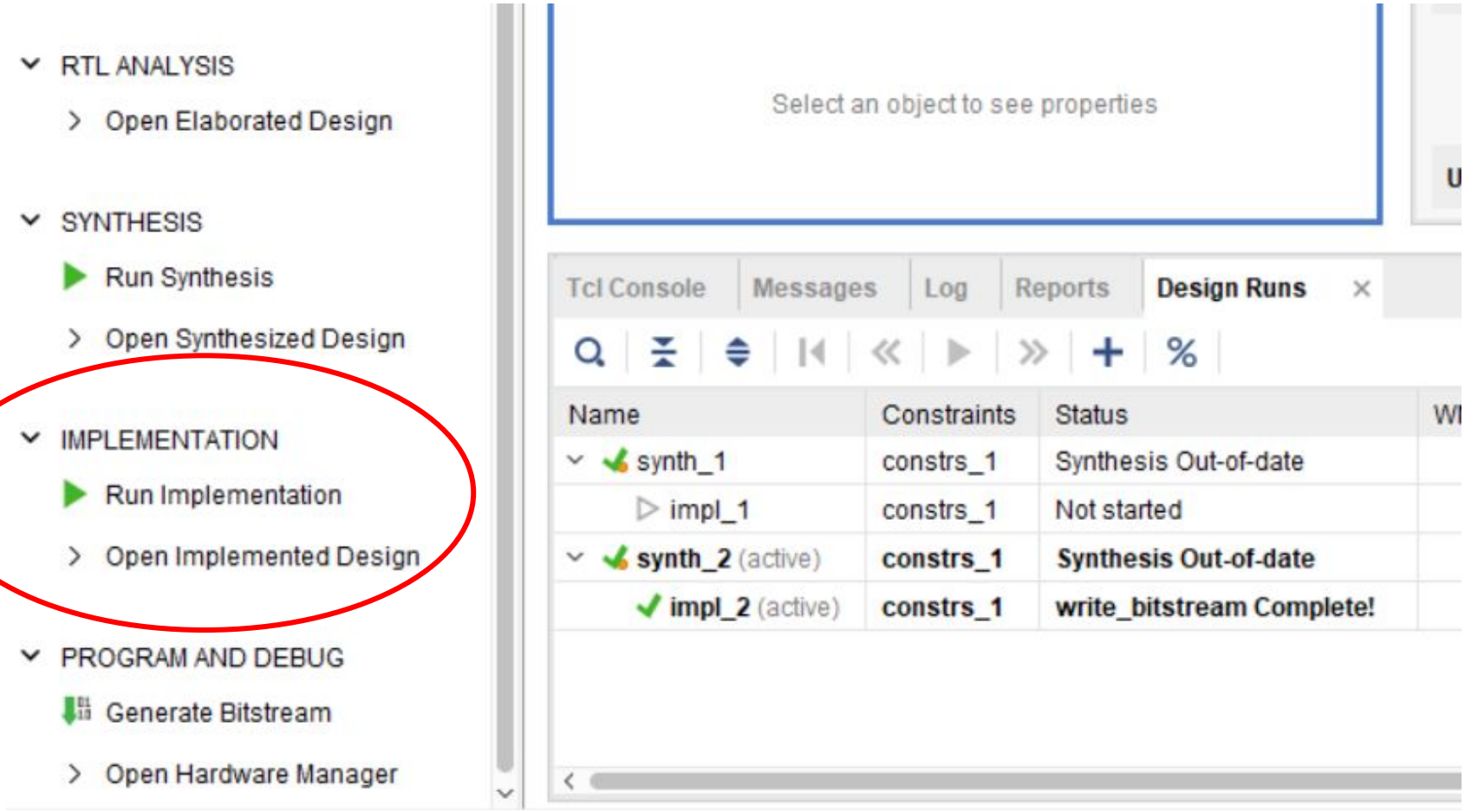
- The design check so far was not device specific.
- Development boards have limited resources. To properly integrate, the correct device must be chosen.
- Choose the desired device (setting/project device/xc7a100tcsg324-1)
- If the device is different change the last name as required.
- At this stage also a **CONSTRAINT FILE** must be created which will map the design module ports to the ports of the actual device. Complete constraint file can be found at

  https://github.com/Digilent/digilent-xdc/blob/master/Nexys-4-Master.xdc

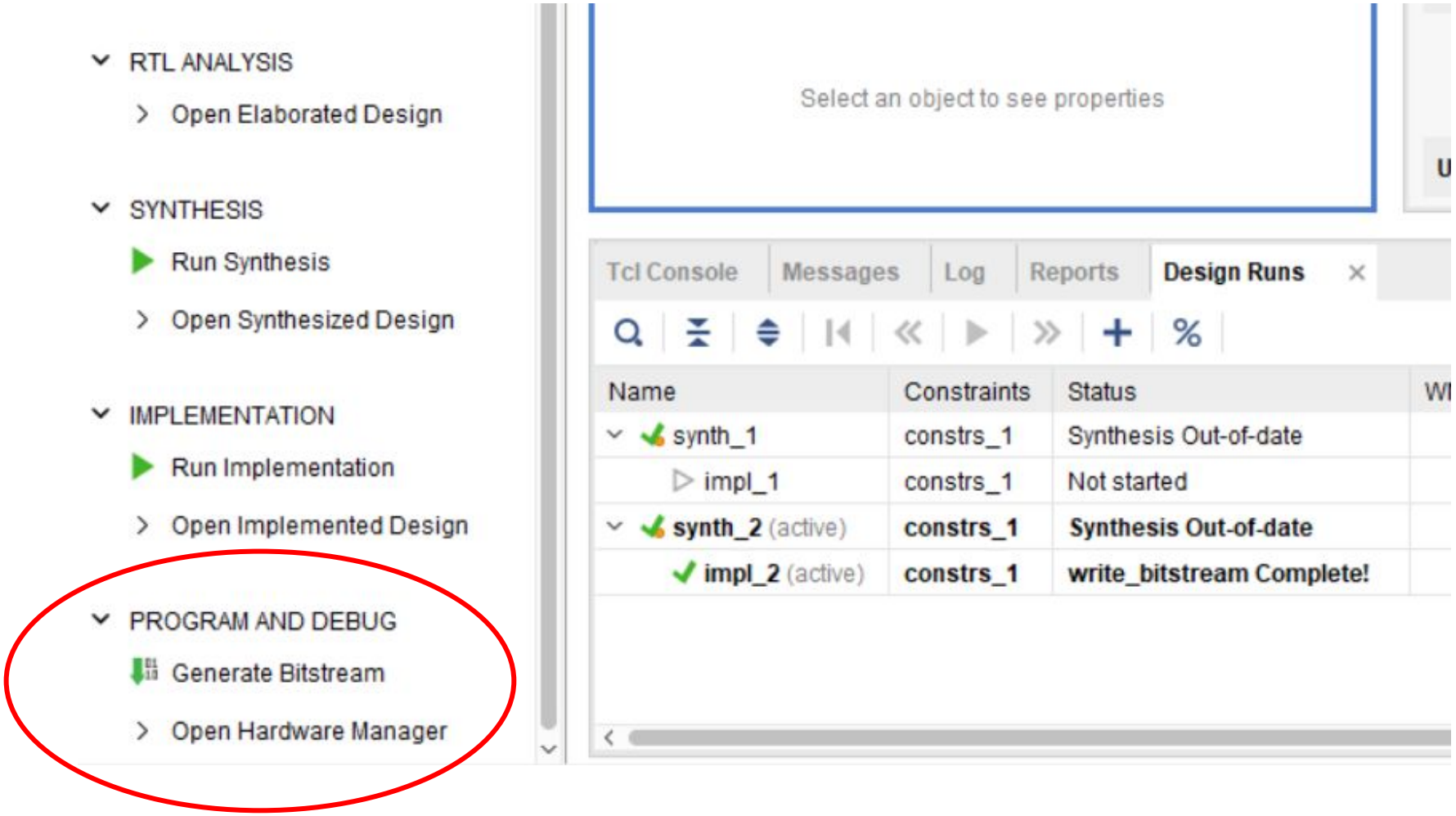
Design Implementation

```
set_property -dict { PACKAGE_PIN J15   IOSTANDARD LVCMOS33 } [get_ports { in0[0] }];
set_property -dict { PACKAGE_PIN L16   IOSTANDARD LVCMOS33 } [get_ports { in0[1] }];
set_property -dict { PACKAGE_PIN M13   IOSTANDARD LVCMOS33 } [get_ports { in0[2] }];
set_property -dict { PACKAGE_PIN R15   IOSTANDARD LVCMOS33 } [get_ports { in0[3] }];
set_property -dict { PACKAGE_PIN R17   IOSTANDARD LVCMOS33 } [get_ports { in1[0] }];
set_property -dict { PACKAGE_PIN T18   IOSTANDARD LVCMOS33 } [get_ports { in1[1] }];
set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports { in1[2] }];
set_property -dict { PACKAGE_PIN R13   IOSTANDARD LVCMOS33 } [get_ports { in1[3] }];
set_property -dict { PACKAGE_PIN T8    IOSTANDARD LVCMOS33 } [get_ports { cin }];
set_property -dict { PACKAGE_PIN H17   IOSTANDARD LVCMOS33 } [get_ports { out[0] }];
set_property -dict { PACKAGE_PIN K15   IOSTANDARD LVCMOS33 } [get_ports { out[1] }];
set_property -dict { PACKAGE_PIN J13   IOSTANDARD LVCMOS33 } [get_ports { out[2] }];
set_property -dict { PACKAGE_PIN N14   IOSTANDARD LVCMOS33 } [get_ports { out[3] }];
set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports { cout }];
```

Design Implementation



Bit stream generation



Upload in FPGA Board

